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# GWPopulation: Hardware agnostic population inference for compact binaries and beyond

**Colm Talbot** [1], **Amanda Farah** [2], **Shanika Galaudage** [3,4], **Jacob Golomb** [5,6], **and Hui Tong** [7,8]

**1** Kavli Institute for Cosmological Physics, University of Chicago, USA **2** Department of Physics, University of Chicago, Chicago, USA **3** Laboratoire Lagrange, Université Côte d'Azur, Observatoire de la Côte d'Azur, CNRS, Bd de l'Observatoire, 06300, France **4** Laboratoire Artemis, Université Côte d'Azur, Observatoire de la Côte d'Azur, CNRS, Bd de l'Observatoire, 06300, France **5** Department of Physics, California Institute of Technology, Pasadena, CA **6** LIGO Laboratory, California Institute of Technology, Pasadena, CA **7** School of Physics and Astronomy, Monash University, VIC 3800, Australia **8** OzGrav: The ARC Centre of Excellence for Gravitational Wave Discovery, Clayton VIC 3800, Australia

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- @sterinaldi

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## Summary

Since the first direct detection of gravitational waves by the LIGO–Virgo collaboration in 2015 (B. P. Abbott & et al., 2016), the size of the gravitational-wave transient catalog has grown to nearly 100 events (R. Abbott & et al., 2023b), with more than as many observed during the ongoing fourth observing run. Extracting astrophysical/cosmological information from these observations is a hierarchical Bayesian inference problem. `GWPopulation` is designed to provide simple-to-use, robust, and extensible tools for hierarchical inference in gravitational-wave astronomy/cosmology. It has been widely adopted for gravitational-wave astrnomy, including producing flagship results for the LIGO-Virgo-KAGRA collaborations (e.g., R. Abbott & et al. (2023a), The LIGO Scientific Collaboration et al. (2024))[1]. While designed to work with observations of compact binary coalescences, `GWPopulation` may be available to a wider range of hierarchical Bayesian inference problems.

Building on `Bilby` (Ashton et al., 2019), `GWPopulation` can easily be used with a range of stochastic samplers through a standard interface. By providing access to a range of array backends (`numpy` (Harris et al., 2020), `JAX` (Bradbury et al., 2018), and `cupy` (Okuta et al., 2017) are currently supported) `GWPopulation` is hardware agnostic and can leverage hardware acceleration to meet the growing computational needs of these analyses. Included in the package are:

- implementations of the most commonly used likelihood functions in the field.
- commonly used models for describing the astrophysical population of merging compact binaries. Including the "PowerLaw+Peak" and "PowerLaw+Spline" mass models, "Default" spin model, and "PowerLaw" redshift models used in the latest LIGO-Virgo-KAGRA collaboration analysis of the compact binary population.
- functionality to simultaneously infer the astrophysical distribution of sources and cosmic expansion history using the "spectral siren" method (Ezquiaga & Holz, 2022).
- a standard specification allowing users to define additional models.

## Statement of need

Hierarchical Bayesian inference is the standard method for inferring parameters describing the astrophysical population of compact binaries and the cosmic expansion history (e.g., Thrane

[1] For a full listing of papers using `GWPopulation`, see the citations for the previous publication.

& Talbot (2019), Vitale et al. (2022)). The first step in the hierarchical inference process is drawing samples from the posterior distributions for the source parameters of each event under a fiducial prior distribution along with a set of simulated signals used to quantify the sensitivity of gravitational-wave searches. Next, these samples are used to estimate the population likelihood using Monte Carlo integration with a computational cost that grows quadratically with the size of the observed population. Since evaluating these Monte Carlo integrals is embarrassingly parallel, this is a prime candidate for hardware acceleration using graphics/tensor processing units. `GWPopulation` provides functionality need to perform this second step.

Maximizing the information we can extract from the gravitational-wave transient catalog requires a framework where potential population models can be quickly constrained with the observed data with minimal boilerplate code. Additionally, the availability of a standard open-source implementation improves the reliability and reproducibility of published results. `GWPopulation` addresses all of these points by providing a standard, open-source, implementation of the standard functionality needed to perform population analyses while enabling user-defined models to be provided by a `Python` function/class definition. The flexible backend system means hardware acceleration can be used with minimal coding effort. Using `GWPopulation` on Google Colab, it is possible to perform an exploratory analysis with a new population model in minutes and produce production-quality results without needing high-performance/throughput computing clusters. With access to high throughput computing resources, a wide range of potential models can be easily explored using the associated `gwpopulation_pipe` (Talbot, 2021) package.

# Related packages

Several other packages are actively used and maintained in the community that can be used for population inference that operate in complementary ways to `GWPopulation`.

- `GWInferno` (Edelman et al., 2023) is a package for hierarchical inference with gravitational-wave sources intended for use with `numpyro` targeting high-dimensional models. `GWInferno` includes many population models initially adapted from `GWPopulation`.
- There are a wide range of packages designed for joint astrophyical and cosmological inference with gravitational-wave transients including `icarogw` (Mastrogiovanni et al., 2024), `gwcosmo` (Gray et al., 2023), `MGCosmoPop` (Mancarella & Genoud-Prachex, 2022), and `CHIMERA` (Borghi et al., 2024). `icarogw` supports some harware acceleration using `cupy` but some cosmological calculations are limited to CPU support only. `chimera` is `JAX`-compatible and supports flat Lambda-CDM cosmologies along with analysis using galaxy catalogs.
- `vamana` (Tiwari, 2021) models the compact binary distribution as a mixture of Gaussians and power-law distributions, `popmodels` (Wysocki & O'Shaughnessy, 2017--) implements a range of parametric models for the compact binary distribution and supports sampling via `emcee` (Foreman-Mackey et al., 2013), neither supports hardware acceleration at the time of writing.

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